



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2006

YEAR 12
TRIAL HIGHER SCHOOL
CERTIFICATE

Mathematics Extension 1

General Instructions

- Working time – 2 Hours
- Reading time – 5 Minutes
- Write using black or blue pen
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work.
- Hand in your answer booklets in 4 sections. Section A (Questions 1 and 2), Section B (Questions 3 and 4), Section C (Questions 5 and 6) and Section D (Question 7).

Total Marks – 84

- Attempt Questions 1 – 7.
- All QUESTIONS are of equal value.

Examiner: *R. Boros*

Section A – Start a new booklet**Marks****Question 1. (12 marks)**

- a)
- (i) Evaluate $\int_0^1 \frac{x}{x^2+1} dx$ leaving your answer in exact form. 2
- (ii) Evaluate $\int_{-2}^{2\sqrt{3}} \frac{1}{x^2+4} dx$ leaving your answer in exact form. 2
- b) Find the gradient of the tangent to the curve $y = \tan^{-1}(\sin x)$ at $x = 0$. 2
- c) Solve for x , $\frac{1}{x+1} < 3$. 2
- d) Give the general solution of the equation, $\cos\left(\theta + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$. 2
- e) If $f(x) = 8x^3$, then find the inverse function $f^{-1}(x)$. 2

Question 2. (12 marks)

- a) Find the co-ordinates of the point P that divides the interval $A(-4, -6)$ and $B(6, -1)$ externally in the ratio 3:1. 2
- b) (i) Sketch the graph of $y = |2x - 4|$. 2
- (ii) Using your graph, or otherwise, solve the inequation $|2x - 4| > x$. 2
- c) Use the substitution $u = 1 + x$ to evaluate, $\int_{-1}^3 x\sqrt{1+x} .dx$. 2
- d) Solve for n , $2 \times {}^nC_4 = 5 \times {}^nC_2$. 2
- e) What is the least distance between the circle $x^2 + y^2 + 2x + 4y = 1$ and the line $3x + 4y = 6$? (Leave your answer in exact form.) 2

End of Section A

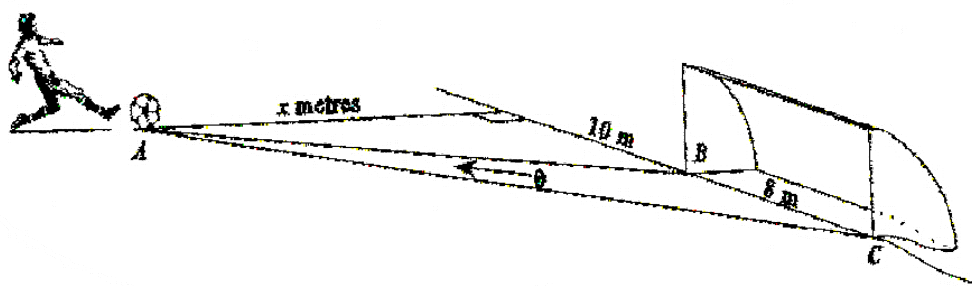
Section B – Start a new booklet**Marks****Question 3. (12 marks)**

- a) If the roots of the equation, $x^4 - 2x^3 - 5x + 1 = 0$, are t_1, t_2, t_3, t_4 ,
find $\sum_{i=1}^4 (t_i t_j t_k)^{-1}$, such that $i \neq j \neq k$. 2
- b) State the domain and range of the function $y = 2 \sin^{-1} \left(\frac{x}{3} \right)$.
Hence sketch the curve. 3
- c) A bowl of water heated to 100°C is placed in a coolroom where the temperature is maintained at -5°C . After t minutes, the temperature $T^\circ \text{C}$ of the water is changing so that $\frac{dT}{dt} = -k(T + 5)$.
- (i) Prove that $T = Ae^{-kt} - 5$ satisfies this equation and find the value of A . 1
- (ii) After 20 minutes, the temperature of the water has fallen to 40°C . How long, to the nearest minute, will the water need to be in the coolroom before ice begins to form, (i.e. the temperature falls to 0°C). 2
- d) (i) Show that the equation $\ln x + x^2 - 4x = 0$ has a root lying between $x = 3$ and $x = 4$. 2
- (ii) By taking $x = 4$ as a first approximation, use one application of Newton's Method to obtain another approximation for the root, to 2 decimal places. Is this newer approximation a better one? Explain. 2

Question 4. (12 marks)**Marks**

- a)** The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. It is given that the chord PQ has equation $y = \left(\frac{p+q}{2}\right)x - apq$.
- (i) Show that the gradient of the tangent at P is p . 1
- (ii) Prove that if PQ passes through the focus, then the tangent at P is parallel to the normal at Q . 2
- b)** A committee of five is to be formed from 4 Liberal senators, 3 Labor senators and 2 Democrat senators.
- (i) How many different committees can be formed that have 3 Liberals, 1 Labor and 1 Democrat? 1
- (ii) If the committee is to be chosen at random, what is the probability that there will be a Liberal majority in the committee? 2
- c)**
- (i) Express $7\cos\theta - \sin\theta$ in the form $R\cos(\theta + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$. 2
- (ii) Hence solve $7\cos\theta - \sin\theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$, giving your answer to the nearest degree. 2
- d)** Find the values of the constants a and b if $x^2 - 2x - 3$ is a factor of the polynomial $P(x) = x^3 - 3x^2 + ax + b$. 2

End of Section B

Section C – Start a new booklet**Marks****Question 5. (12 marks)****a)**

A soccer player A is x metres from a goal line of a soccer field. He takes a shot at the goal BC , with the ball not leaving the ground.

- (i) Show that the angle θ within which he must shoot is given by

$$\theta = \tan^{-1} \left(\frac{8x}{180 + x^2} \right) \text{ when he is 10 metres to one side of the near}$$

goal post and 18 metres to the same side of the far post.

2

- (ii) Find the value of x which makes this angle a maximum. (Leave your answer in exact form).

2

- b)** A particle moves in a straight line such that its velocity V m/s is given by

$V = 2\sqrt{2x-1}$ when it is x metres from the origin. If $x = \frac{1}{2}$ when $t = 0$ find:

- (i) the acceleration.

1

- (ii) an expression for x in terms of t .

2

c)

Find the volume of the solid obtained by rotating $y = \sin^{-1} x$ about the y -axis

between $y = -\frac{\pi}{4}$ and $y = \frac{\pi}{4}$. Answer in exact form.

3

d)

The perimeter of a circle is increasing at 3 cm/s. Leaving your answer in terms of π , find the rate at which the area is increasing when the perimeter is 1m.

2

Question 6. (12 marks)**Marks****a)**

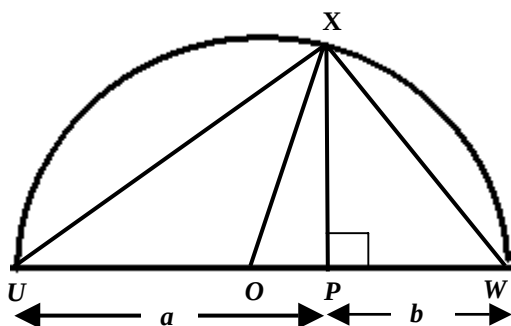
Consider the following three expressions involving n , where n is a positive integer:

$$5^n + 3, 7^n + 5, 5^n + 7$$

- (i) By substituting values of n , show that $7^n + 5$ is the only one of these expressions which could be divisible by 6 for all positive integers n . 1
- (ii) Use mathematical induction to show that the expression $7^n + 5$ is in fact divisible by 6 for all positive integers n . 2

b)

Not to scale



In the diagram UXW is a semi-circle with O as a midpoint of diameter UW . The point P lies on UW and XP is perpendicular to UW . The length of $UP = a$ units and $PW = b$ units are shown.

- (i) Explain why $OX = \frac{a+b}{2}$. 1
- (ii) Show that $\angle UXP = \angle XWP$. 1
- (iii) Deduce that $XP = \sqrt{ab}$. 1
- (iv) By using the diagram show that $\frac{a+b}{2} \geq \sqrt{ab}$. 1

c)

The displacement x metres of a particle from the origin is given by

$$x = 5 \cos\left(3t - \frac{\pi}{6}\right), \text{ where } t \text{ is the time lapsed in seconds.}$$

- (i) Show that $\ddot{x} = -9x$. 1
- (ii) Find the period of the motion 1

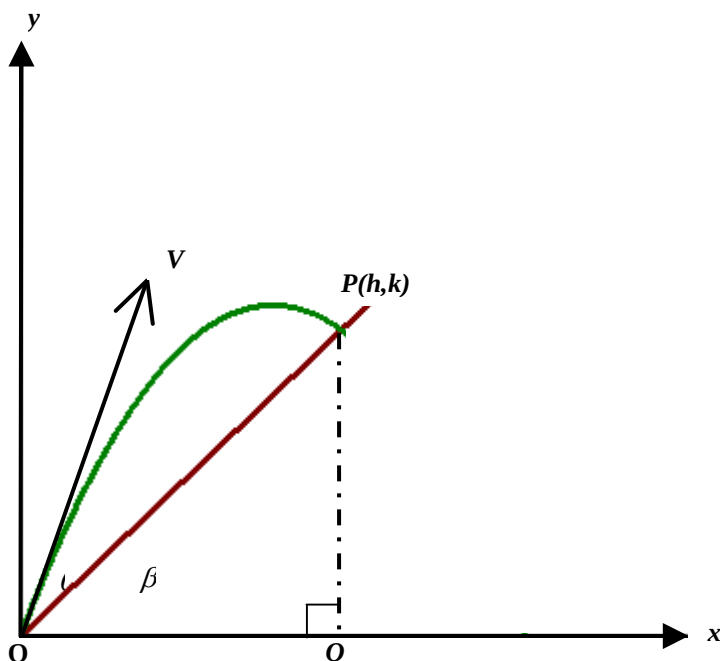
Marks

d) Suppose that $(5 + 2x)^{12} = \sum_{k=0}^{12} a_k x^k$.

(i) Use the binomial theorem to write the expression for a_k . 1

(ii) Show that $\frac{a_{k+1}}{a_k} = \frac{24 - 2k}{5k + 5}$ 2

End of Section C

Section D – Start a new booklet**Marks****Question 7. (12 marks)**

A projectile is fired from the origin with a velocity V and an angle of elevation θ , where $\theta \neq 90^\circ$. You may assume that $x = Vt \cos \theta$ and $y = -\frac{1}{2}gt^2 + Vt \sin \theta$, where x and y are the horizontal and vertical displacements of the projectile in metres from O at time t seconds after firing, and g is the acceleration due to gravity.

- (i) Show that the Cartesian equation of the flight of the projectile is:

$$y = x \tan \theta - \frac{g}{2V^2 \cos^2 \theta} x^2 \quad 1$$

- (ii) Suppose the projectile is fired up a plane inclined at β to the horizontal so that $0^\circ \leq \beta \leq \theta$. If the projectile strikes the plane at $P(h, k)$, show that:

$$h = \frac{(\tan \theta - \tan \beta) 2V^2 \cos^2 \theta}{g} \quad 2$$

- (iii) Hence, show that the range OP of the projectile can be given by

$$OP = \frac{2V^2 \sin(\theta - \beta) \cos \theta}{g \cos^2 \beta} \quad 4$$

Marks

- (iv) Given the fact that $2 \sin(x - \beta) \cos x = \sin(2x - \beta) - \sin \beta$. Show that the maximum value of the range of OP is given by:

$$\frac{V^2}{g(1 + \sin \beta)} \quad 4$$

- (v) If the angle of inclination of the plane is 14° , at what angle to the horizontal should the projectile be fired in order to attain maximum range? 1

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

Question 1:

$$\begin{aligned} \text{a) i)} \int_0^1 \frac{x}{x^2+1} dx &= \frac{1}{2} \int_0^1 \frac{2x}{x^2+1} dx \quad \frac{1}{2} \\ &= \frac{1}{2} \left[\ln(x^2+1) \right]_0^1 \quad \frac{1}{2} \\ &= \frac{1}{2} [\ln 2 - \ln 1] \\ &= \frac{1}{2} \ln 2 \quad 1 \end{aligned}$$

$$\begin{aligned} \text{ii)} \int_{-2}^{2\sqrt{3}} \frac{1}{x^2+4} dx &= \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_{-2}^{2\sqrt{3}} \quad \frac{1}{2} \\ &= \left(\frac{1}{2} \tan^{-1} \frac{2\sqrt{3}}{2} \right) - \left(\frac{1}{2} \tan^{-1} \frac{-2}{2} \right) \\ &= \left(\frac{1}{2} \tan^{-1} \sqrt{3} \right) - \left(\frac{1}{2} \tan^{-1} (-1) \right) \\ &= \left(\frac{1}{2} \cdot \frac{\pi}{3} \right) - \left(\frac{1}{2} \cdot \frac{-\pi}{4} \right) \quad \frac{1}{2} \\ &= \frac{\pi}{6} - - \frac{\pi}{8} \\ &= \frac{7\pi}{24} \quad 1 \end{aligned}$$

b) Find the gradient of the tangent to the curve $y = \tan^{-1}(\sin x)$ at $x = 0$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{1 + (\sin x)^2} \times (\cos x) \quad -1 \text{ for no } x \cos x. \\ &= \frac{\cos x}{1 + \sin^2 x} \quad 1 \end{aligned}$$

when $x = 0$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\cos 0}{1 + \sin^2 0} \\ &= \frac{1}{1} \end{aligned}$$

$$= 1$$

c) solve for x , $\frac{1}{x+1} < 3$

$$x + 1 \neq 0$$

$$\therefore x \neq -1 \quad \frac{1}{2}$$

$$\times \text{ by } (x+1)^2$$

$$\frac{(x+1)^2}{x+1} < 3(x+1)^2 \quad \frac{1}{2}$$

$$x+1 < 3x^2 + 6x + 3$$

$$0 < 3x^2 + 6x + 3 - (x+1)$$

$$0 < 3x^2 + 5x + 2$$

$$\frac{(3x+3)(3x+2)}{3}$$

$$0 < (x+1)(3x+2) \quad \frac{1}{2}$$

$$= \frac{3(x+1)(3x+2)}{3}$$

$$x < -1, \quad x > -2/3 \quad \frac{1}{2}$$

d) General solution for

$$\cos(\theta + \pi/4) = 1/\sqrt{2}$$

$$\cos(\theta + \pi/4) = \cos \pi/4 \quad \frac{1}{2}$$

$$\theta + \pi/4 = 2n\pi \pm \pi/4 \quad 1$$

$$\theta = 2n\pi \pm \pi/4 - \pi/4 \quad \frac{1}{2}$$

e) $f(x) = 8x^3$ find inverse function $f^{-1}(x)$

$$f(f^{-1}(x)) = x = f^{-1}(f(x)) \quad \frac{1}{2}$$

$$f(f^{-1}(x)) = x$$

$$\therefore 8(f^{-1}(x))^3 = x$$

$$f^{-1}(f(x)) = \sqrt[3]{8x^3}$$

$$f^{-1}(x)^3 = x/8$$

$$\therefore f^{-1}(x) = \sqrt[3]{\frac{x}{8}}$$

$$= \frac{2x}{2}$$

$$= \frac{\sqrt[3]{x}}{2} \quad \frac{1}{2}$$

$$= x \quad \frac{1}{2}$$

~~$$f^{-1}(x) = \frac{\sqrt[3]{8x^3}}{2}$$~~

$$\therefore f^{-1}(x) = \frac{\sqrt[3]{8x^3}}{2} \quad \frac{1}{2}$$

~~$$f^{-1}(x) = \frac{\sqrt[3]{8x^3}}{2}$$~~

$$-1 \text{ for } \sqrt[3]{8} = 2\sqrt{2}$$

$$-1 \text{ for } \frac{\sqrt[3]{x}}{8}$$

Question 2.

a) A (-4, -6) B (6, -1) divide externally in ratio 3:1

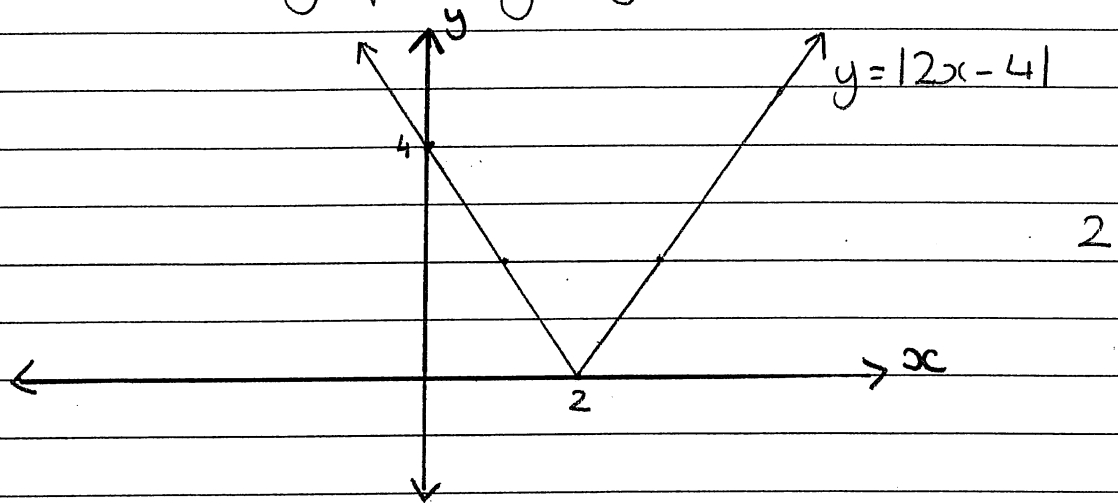
$$x\text{-co-ordinate} = \frac{(3 \times 6) - (1 \times -4)}{3 - 1} = 11 \frac{1}{2}$$

$$y\text{-co-ordinate} = \frac{(3 \times -1) - (1 \times -6)}{3 - 1} = \frac{3}{2} \frac{1}{2}$$

$\therefore$ The co-ordinates of P are $(11, \frac{3}{2})$

$\frac{1}{2}$ if x_2, x_1, y_2, y_1 are switched.

b) i) sketch the graph of $y = |2x - 4|$



ii) solve $|2x - 4| > x$

$$\sqrt{(2x - 4)^2} > x$$

$$(2x - 4)^2 > x^2$$

$$4x^2 - 16x + 16 > x^2$$

$$3x^2 - 16x + 16 > 0$$

$$(3x - 12)(3x - 4) > 0$$

3

$$3(x - 4)(3x - 4) > 0$$

3

$$x > 4, x < \frac{4}{3}$$

2

c) Use $u = 1+x$ to evaluate $\int_{-1}^3 x \sqrt{1+x} dx$

$$u = 1+x$$

limits:

$$\therefore \frac{du}{dx} = 1 \quad \therefore x = u-1 \quad \begin{array}{l} x=3 \therefore u=1+3 \Rightarrow u=4 \\ x=-1 \therefore u=1-1 \Rightarrow u=0 \end{array}$$

$$du = dx$$

$$\int_0^4 (u-1) \cdot \sqrt{u} du \quad 1/2$$

$$= \int_0^4 (u-1) \cdot u^{1/2} du$$

$$= \int_0^4 (u^{3/2} - u^{1/2}) du$$

$$= \left[\frac{u^{5/2}}{5/2} - \frac{u^{3/2}}{3/2} \right]_0^4 \quad 1/2$$

$$= \left[\frac{2 \cdot 4^{5/2}}{5} - \frac{2 \cdot 4^{3/2}}{3} \right] - \left[\frac{2 \cdot 0^{5/2}}{5} - \frac{2 \cdot 0^{3/2}}{3} \right]$$

$$= \left[\frac{2 \times 32}{5} - \frac{2 \times 8}{3} \right]$$

$$= \frac{64}{5} - \frac{16}{3}$$

$$= \frac{112}{15} \text{ OR } 7\frac{7}{15} \quad 1/2$$

d) solve for n , $2 \times {}^nC_4 = 5 \times {}^nC_2$ $-1/2$ for $n=-3$

$$\frac{2 \times n!}{(n-4)! 4!} = \frac{5 \times n!}{(n-2)! 2!}$$

$$\div \text{ by } n! \quad \frac{2 \cdot 1}{24 (n-4)!} = \frac{5 \cdot 1}{2 (n-2)!}$$

$$\times \text{ by } (n-4)! \quad \frac{2}{24} = \frac{5}{2} \cdot \frac{1}{(n-2)(n-3)}$$

$$\times \text{ by } 2/5 \quad \frac{1}{30} = \frac{1}{(n-2)(n-3)}$$

check:

$$2 \times {}^8C_4 = 140 = 5 \times {}^8C_2$$

$$\therefore n = 8. \quad 1$$



2e) circle $x^2 + y^2 + 2x + 4y = 1$

$$\therefore (x+1)^2 + (y+2)^2 = 6$$

$$\therefore \text{centre } (-1, -2) \quad \text{radius } \sqrt{6} \quad \frac{1}{2}$$

$$\text{line } 3x + 4y = 6 \Rightarrow 3x + 4y - 6 = 0$$

least distance between circle & line is
the distance between the line & centre of the circle less the radius.

$$d = \frac{|(3 \times -1) + (4 \times -2) - 6|}{\sqrt{3^2 + 4^2}}$$

$$= \frac{|-3 - 8 - 6|}{5}$$

$$= \frac{|-17|}{5} \quad \frac{1}{2}$$

$$\therefore \text{minimum distance} = \frac{17}{5} - \sqrt{6} \quad \frac{1}{2}$$

17/5 1 mark only.

SECTION B QUESTION 3

$$a) \sum_{\substack{i=1 \\ i \neq j \neq k}}^4 (t_i t_j t_k)^{-1} = \frac{1}{t_1 t_2 t_3} + \frac{1}{t_1 t_2 t_4} + \frac{1}{t_1 t_3 t_4} + \frac{1}{t_2 t_3 t_4}$$

$$= \frac{t_1 + t_2 + t_3 + t_4}{t_1 t_2 t_3 t_4}$$

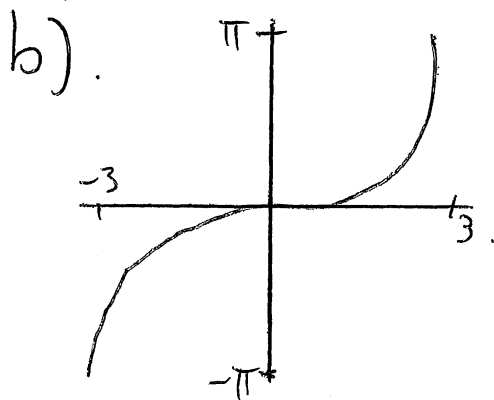
$$t_1 + t_2 + t_3 + t_4 = -\frac{b}{a}$$

$$= 2$$

$$t_1 t_2 t_3 t_4 = \frac{c}{a}$$

$$= 1$$

$$\text{So } \sum_{\substack{i=1 \\ i \neq j \neq k}}^4 (t_i t_j t_k)^{-1} = 2$$



Domain: $-1 \leq \frac{x}{3} \leq 1$
 $-3 \leq x \leq 3$

Range: $-\frac{\pi}{2} \leq \sin^{-1}\left(\frac{x}{3}\right) \leq \frac{\pi}{2}$

$$-\pi \leq 2 \sin^{-1}\left(\frac{x}{3}\right) \leq \pi$$

$$-\pi \leq y \leq \pi$$

$$c i) \quad LHS = \frac{dT}{dt}$$

$$= -kAe^{-kt}$$

$$RHS = -k(T+S)$$

$$\begin{aligned} &= -k(Ae^{-kt} - 5 + 5) \\ &= -kAe^{-kt} \\ &= LHS. \end{aligned}$$

Initial conditions

$$\begin{aligned} 100 &= Ae^{-k0} - 5 \\ A &= 105. \end{aligned}$$

ii) After 20 minutes

$$40 = 105e^{-20k} - 5$$

$$\frac{45}{105} = e^{-20k}$$

$$-20k = \ln \frac{3}{7}$$

$$k = \frac{\ln \frac{3}{7}}{-20}$$

At $0^\circ C$.

$$0 = 105e^{-kt} - 5$$

$$e^{-kt} = \frac{5}{105}$$

$$t = \frac{\ln \frac{1}{21}}{-k}$$

$$t = 72 \text{ minutes.}$$

di) Since $f(x) = \ln x + x^2 - 4x$ is a continuous function and

$$f(3) = \ln 3 + 3^2 - 4 \times 3 \\ \approx -1.9 < 0$$

and

$$f(4) = \ln 4 + 4^2 - 4 \times 4 \\ \approx 1.4 > 0$$

Therefore $f(x) = \ln x + x^2 - 4x$ must have a root between $x=3, x=4$.

$$ii) f'(x) = \frac{1}{x} + 2x - 4.$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_0 = 4$$

$$x_1 = 4 - \frac{\ln 4 + 16 - 16}{\frac{1}{4} + 8 - 4}$$

$$\approx 3.67.$$

Yes, since we know $f(x)$ has a root between 3 and 4 and this approximation is closer to 3 than the first approximation of 4.

QUESTION 4.

i). $y = \frac{x^2}{4a}$

$$\frac{dy}{dx} = \frac{x}{2a}.$$

At $P(2ap, ap^2)$

$$m_p = \frac{2ap}{2a}$$

$$= p.$$

ii) Similarly to part (i) the tangent at Q is $m_q = q$.

Thus the gradient of the normal will be $m = -\frac{1}{q}$.

Given that the chord goes through the Locus $(0, a)$. then

$$a = \left(\frac{p+q}{2}\right)0 - apq.$$

$$pq = -1$$

$$\therefore q = -\frac{1}{p}.$$

Thus the gradient of the normal at Q will be $-\frac{1}{q} = -\left(-\frac{1}{p}\right)$

$$= p.$$

∴ Tangent at P is parallel to the normal at Q.

$$b). i) \binom{4}{3} \binom{3}{1} \binom{2}{1} = 24.$$

$$ii) n(\text{Sample space}) = \binom{9}{5} = 126.$$

$$\begin{aligned} n(E) &= n(4 \text{ l.b}) + n(3 \text{ l.b}) \\ &= \binom{4}{4} \binom{5}{1} + \binom{4}{3} \binom{5}{2} \\ &= 5 + 40 \\ &= 45. \end{aligned}$$

$$\begin{aligned} P(E) &= \frac{n(E)}{n(S)} \\ &= \frac{45}{126} \\ &= \frac{5}{14}. \end{aligned}$$

$$c) R = \frac{\sqrt{49+1}}{5\sqrt{2}}$$

$$\tan \alpha = \frac{1}{7}$$

$$\alpha = 8^{\circ} 8'$$

$$\text{So } 7\cos\theta - \sin\theta = 5\sqrt{2} \cos(\theta + 8^{\circ} 8').$$

$$ii) 5\sqrt{2} \cos(\theta + 88^\circ) = 5$$

$$\cos(\theta + 88^\circ) = \frac{1}{\sqrt{2}}$$

$$\theta + 88^\circ = 45^\circ, 315^\circ$$

$$\theta \approx 37^\circ, 307^\circ$$

$$d) p(-1) = (-1)^3 - 3(-1)^2 + a(-1) + b$$

$$= -4 - a + b$$

$$p(3) = (3)^3 - 3(3)^2 + (3)a + b$$

$$= 3a + b$$

$$\text{So } -4 - a + b = 0$$

$$\text{and } 3a + b = 0$$

Ⓐ
Ⓑ

$$\text{from } \textcircled{A} \quad a = b - 4$$

sub into Ⓑ

$$3b - 12 + b = 0$$

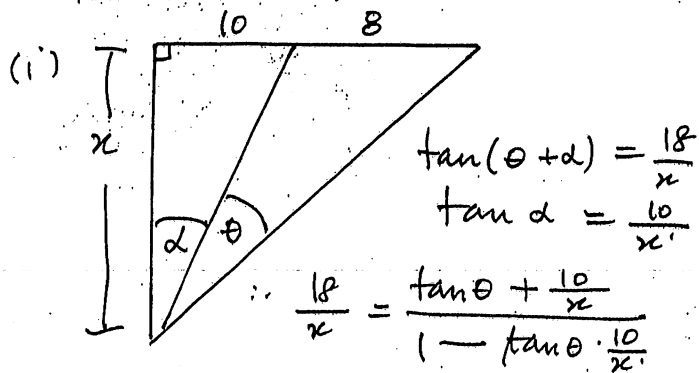
$$b = 3$$

$$a = -1$$

so

$$\therefore a = -1, b = 3$$

Question (5)



$$(\tan \theta)x^2 - 8x + 180 \tan \theta = 0$$

$$\tan \theta = \frac{8x}{180 + x^2}$$

$$\therefore \theta = \tan^{-1} \left(\frac{8x}{180 + x^2} \right)$$

$$\frac{d\theta}{dx} = \frac{(180 + x^2)8 - 8x(2x)}{(x^2 + 180)^2}$$

$$1. \quad \frac{1440 - 8x^2}{(x^2 + 180)^2} = 0$$

$$8(x^2 - 180) = 0, \quad x^2 = 180$$

$$x = 6\sqrt{5}$$

Test:

θ	13	$6\sqrt{5}$	14
$\frac{d\theta}{dx}$	+	0	-1568

①

(b) $v = 2(2x - 1)^{\frac{1}{2}}$

(i) $v^2/2 = 4x - 2$

$$\dot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4$$

[1]
①

(ii) $\frac{dx}{dt} = 2(2x - 1)^{\frac{1}{2}}$

$$\frac{dx}{2x - 1} = \frac{1}{2} dt$$

$$\therefore t = \frac{1}{2} (2x - 1)^{\frac{1}{2}} + C$$

When $t = 0$, $x = \frac{1}{2} \Rightarrow C = 0$

$$\therefore 2t = \sqrt{2x - 1}$$

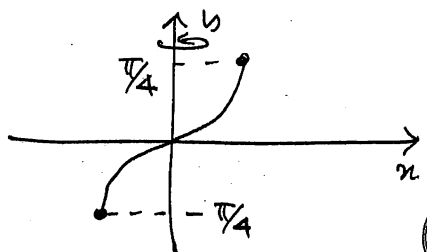
$$1 + 4t^2 = 2x$$

$$\therefore x = \frac{1 + 4t^2}{2}$$

$$= 2t^2 + \frac{1}{2}$$

[2]
②

(c)



$$\left(\sin^2 y = \frac{1 - \cos 2y}{2} \right)$$

$$V = 2\pi \int_0^{\pi/4} \sin^2 y \, dy$$

$$= \pi \int_0^{\pi/4} (1 - \cos 2y) \, dy$$

$$= \pi \left[y - \frac{\sin 2y}{2} \right]_0^{\pi/4}$$

$$= \pi \left(\frac{\pi}{4} - \frac{1}{2} \right) \quad [3]$$

(d)

$$\frac{dp}{dt} = 3, \quad p = 2\pi r$$

$$\therefore t = \frac{p}{2\pi}$$

$$A = \pi \left(\frac{p^2}{4\pi^2} \right) = \frac{p^2}{4\pi}$$

$$\frac{dA}{dt} = \frac{dA}{dp} \cdot \frac{dp}{dt} = \frac{p}{2\pi} \times 3$$

$$p = 100$$

$$\therefore \frac{dA}{dt} = \frac{150}{\pi} \text{ cm}^2/\text{s}$$

$$= \frac{0.015}{\pi} \text{ m}^2/\text{s} \quad [2]$$

12

②

Question (6)

6(a) $n = 1, 6 \nmid 8$

(i) [1]

(ii) For $n = 1$, $7 + 5 = 12$
and $6 \mid 12$

Assume $S(k)$ is true

i.e. $7^5 + 5 = 6M$, $M \in \mathbb{N}$

Consider $n = k + 1$

$$7^{k+1} + 5$$

$$= 7 \cdot 7^k + 5$$

$$= 7(7^5 + 5) - 30$$

$$= 42M - 30 = 6(7M - 5)$$

$$= 6N \text{ where } N = 7M - 5 \in \mathbb{N}$$

$\therefore S(k+1)$ is true when $S(k)$

is true and $S(1)$ is true

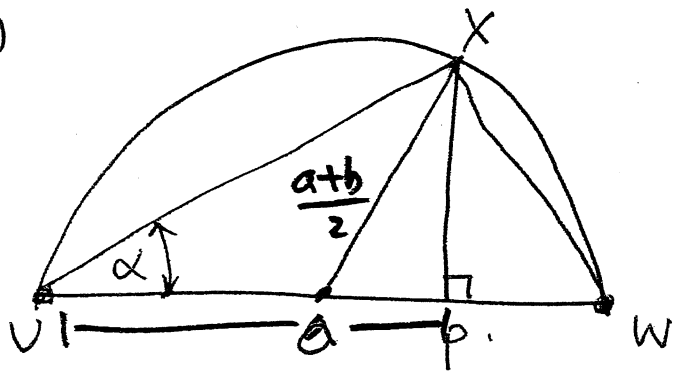
$\therefore$ by mathematical induction

$$6 \mid (7^n + 5) \quad \forall n \in \mathbb{N}$$

②

[2]

(b)



$OX = \frac{1}{2} UW = a + b$.
(radius = $\frac{1}{2}$ diameter)

$$\therefore OX = \frac{a+b}{2} \quad \text{--- (1)} \quad [1]$$

(ii) Let $\angle XUP = \alpha$.

$\angle UXW = 90^\circ$ (Angle in a semicircle)

$$\therefore \angle XWP = 90^\circ - \alpha.$$

In $\triangle UXP$, $\triangle XWP$.

$$\angle XWP = 90 - \alpha \Rightarrow \angle XWP = \alpha$$

(Angle sum of $\triangle XPW$.)

$$\therefore \triangle UXP \parallel \triangle XWP \quad [1]$$

(Equiangular)

In $\triangle XPU$

$$XP^2 + a^2 = XU^2$$

$$\frac{XP}{PW} = \frac{UP}{XP}$$

$$\therefore XP^2 = a \cdot b$$

In $\triangle XWP$

$$XP^2 + b^2 = XW^2.$$

In $\triangle UXW$

$$(a+b)^2 = XU^2 + XW^2$$

$$a^2 + b^2 + 2ab = (a^2 + b^2) + 2XP^2$$

$$\therefore XP^2 = ab \quad [1]$$

$$\Rightarrow XP = \sqrt{ab}.$$

In $\triangle OXP$ $OX \perp XP$
(OX is the hypotenuse) $[1]$

$$\therefore \frac{a+b}{2} > \sqrt{ab}.$$

(c) $\ddot{x} = -15 \sin(3t - \frac{\pi}{6})$

(i) $\ddot{x} = -45 \cos(3t - \frac{\pi}{6})$
 $= -9 [5 \cos(3t - \frac{\pi}{6})]$
 $= -9x \quad [1]$

(ii) $T = \frac{2\pi}{\omega} = \frac{2\pi}{3} \quad [1]$

(d) (i) $a_k = \binom{12}{k} 5^{12-k} 2^k \quad [1]$

(ii) $\frac{a_{k+1}}{a_k} = \frac{\binom{12}{k+1} 5^{11-k} 2^{k+1}}{\binom{12}{k} 5^{12-k} 2^k} = \frac{2}{5} \frac{(12-k)}{k+1} \quad [2]$

QUESTION 7

$$(i) t = \frac{x}{v \cos \theta}$$

$$y = -\frac{gx^2}{2v^2 \cos^2 \theta} + \frac{vx \sin \theta}{v \cos \theta}$$

$$y = x \tan \theta - \frac{gx^2}{2v^2 \cos^2 \theta}$$

(ii)

At P, $y = k = h \tan \theta$, $x = h$

$$h \tan \beta = h \tan \theta - \frac{gh^2}{2v^2 \cos^2 \theta} \text{ from (i)}$$

$$\frac{gh^2}{2v^2 \cos^2 \theta} = h(\tan \theta - \tan \beta)$$

$$h = \frac{(\tan \theta - \tan \beta) 2v^2 \cos^2 \theta}{g}$$

(iii)

$$OP = \frac{h}{\cos \beta}$$

$$= \frac{(\tan \theta - \tan \beta) 2v^2 \cos^2 \theta}{g \cos \beta} \text{ [from (ii)]}$$

$$= \frac{\left(\frac{\sin \theta}{\cos \theta} - \frac{\sin \beta}{\cos \beta} \right) 2v^2 \cos^2 \theta}{g \cos \beta}$$

$$= \frac{(\sin \theta \cos \beta - \sin \beta \cos \theta) 2v^2 \cos \theta}{g \cos^2 \beta}$$

$$= \frac{2v^2 \sin(\theta - \beta) \cos \theta}{g \cos^2 \beta}$$

(iv)

$$OP = \frac{[\sin(2\theta - \beta) - \sin \beta] v^2}{g \cos^2 \beta} \text{ (given)}$$

$$\frac{d(OP)}{(d\theta)} = \frac{2v^2}{g \cos^2 \beta} [2 \cos(2\theta - \beta)]$$

$$OP \text{ max/min } \cos(2\theta - \beta) = 0$$

$$2\theta - \beta = 90^\circ$$

$$\theta = \frac{90^\circ + \beta}{2}$$

$$OP'' = \frac{4v^2}{g \cos^2 \beta} \times -2 \sin(2\theta - \beta)$$

always < 0 as $(2\theta - \beta) < 180^\circ$

$$\therefore \text{max val OP when } \theta = \frac{90^\circ + \beta}{2}$$

$$\text{Max val. OP} = \frac{v^2 (\sin 90^\circ - \sin \beta)}{g(1 - \sin^2 \beta)}$$

$$= \frac{v^2 (1 - \sin \beta)}{g(1 - \sin^2 \beta)}$$

$$= \frac{v^2}{g(1 + \sin \beta)}$$

(v)

$$\text{max val OP when } \theta = \frac{90^\circ + \beta}{2} \text{ [from (iv)]}$$

$$\theta = \frac{90^\circ + 14^\circ}{2}$$

$$\theta = 52^\circ$$